<u>SML Assignment 6</u>

Implementation of Decision Tree for Classification

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Aim: To implement and understand the Decision Tree algorithm for classification tasks and evaluate its performance on a given dataset.

Objective: Students will learn

- Understand the decision tree classification mechanism.
- Implement the decision tree algorithm on a dataset for classification.
- Evaluate the performance using relevant metrics such as accuracy, precision, recall, and F1-score.
- Visualize the decision tree structure and its decision boundaries.

Problem statement: Use the given dataset(s) to demonstrate the application of the Decision Tree algorithm for classification. The task is to classify the data points into different classes based on the features and to understand how the decision tree algorithm splits the data at each node.

Theory:

Decision Tree:

A Decision Tree is a **supervised machine learning algorithm** used for classification and regression tasks.

It works by:

- Splitting data based on feature values
- Creating a tree-like structure of decisions
- Assigning class labels at leaf nodes

2. Mathematical Formula:

Decision Trees decide splits using Impurity Measures.

i). Entropy:

Entropy measures disorder in the data.

$$E(S) = \sum_{i=1}^{c} - p_i \log_2 p_i$$

Where:

pi = probability of class i

High entropy → Mixed classes

Low entropy → Pure node

ii.)Information Gain:

Information Gain tells us how good a split is.

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

Decision Tree chooses the split with maximum Information Gain

iii.) Gini Impurity:

Gini measures probability of incorrect classification.

$$Gini = 1 - \sum_{i=1}^{n} (p_i)^2$$

Lower Gini → Better split

3. Importance of Decision Tree

Decision Trees are important because:

- Easy to interpret
- No need for feature scaling
- Handles numerical & categorical data
- Captures non-linear relationships
- Works well with imbalanced datasets (using class_weight)
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4. Applications

Decision Trees are widely used in:

- Medical diagnosis
- Fraud detection
- Intrusion Detection Systems
- Customer churn prediction
- Credit risk analysis
- Spam detection

5. Performance Metrics

Accuracy: Measures proportion of correctly classified samples.

Precision: Measures correctness of positive predictions.

Recall: Measures ability to detect actual positives.

F1-Score: Harmonic mean of precision and recall.

Confusion Matrix: Shows TP, TN, FP, FN values.

Cross-Validation: Technique used to evaluate model performance by splitting data into multiple folds.

6. Dataset Description

Dataset Used: iPhone Purchase Records

- Dependent variable: Purchase (whether the user bought the iPhone)
- Independent variables: Age, Salary, Gender, etc.
- Dataset contains: Customer demographic data used to predict purchasing behavior.

Dataset Used: Bank Marketing Dataset

- Dependent variable: y (whether the client subscribed to a term deposit)
- Independent variables: Age, Job, Marital Status, Education, Balance, Loan, Contact Type, Duration, Campaign, etc.
- Dataset contains: Customer and campaign-related attributes used for marketing response prediction.

Dataset Used: Medical Risk Classification

- Dependent variable: Risk Level
- Independent variables: Medical and physiological attributes.
- Dataset contains: Patient health indicators used to classify medical risk levels

Dataset Used: Multiclass Sensor Fault Classification

- Dependent variable: Fault Type
- Independent variables: Sensor readings and operational parameters.
- Dataset contains: Machine sensor data used to classify different fault conditions.

Methodology

Input:

- Dataset containing feature variables and target class
- Value of max_depth selected through hyperparameter tuning

Output:

- Predicted class labels (classification)
- Accuracy, Precision, Recall, F1-score
- Confusion Matrix
- Precision-Recall Curve
- Graphs showing performance vs tree depth

Conclusion

Impact of Small max_depth

- High bias
- Low variance
- Simpler model
- May underfit data
- Limited decision boundaries

Impact of Large max_depth

- Low bias
- High variance
- Complex model
- May overfit data
- Captures detailed patterns